

Rootine

Rootine is a to-do list application designed to help users efficiently plan their day, month, and year through manageable and well-defined tasks. It is especially beneficial for individuals who struggle with completing small daily tasks, as Rootine offers a structured approach to planning and achieving goals.

The name Rootine is derived from Root and Routine, giving the impression that you can improve your standing point (the roots).

How to Use Rootine

Task Creation: Users can create tasks with a title, description, and a goal time, which indicates the remaining time to complete the task.

Subtasks: Each task can include multiple subtasks, enabling users to break down larger tasks into more manageable steps. As subtasks are completed, users can “tick them off” and get motivation as they work through the list.

Importance Levels: Assigning an importance level to tasks prioritizes them, placing the most critical tasks at the top of the list and highlighting them in a distinct colour. This ensures that users focus on their most significant tasks.

Example Usage:

Consider planning a productive day with the task "Have a Productive Day." This task might include subtasks such as brushing teeth, taking a shower, eating a healthy breakfast, and tidying the room, each with its own goal time. This methodical approach not only helps in organizing the day but also provides a sense of achievement as each subtask is completed.

Rootine simplifies the process of task management, making it easier for users to stay organized and focused on their daily objectives.

2. Design Rationale

Development Environment: I chose to use the latest version of Android Studio, Google's recommended Integrated Development Environment (IDE). Because this meant I had access to the most advanced tools and features available. This choice was driven by a desire to challenge and enhance my understanding and skills in app development. I decided to use Kotlin with Jetpack Compose

Database Implementation: My app uses a Room database. I designed the database with two tables to store detailed information about tasks and their respective subtasks. This structure allows future improvements, such as tracking task completion times and analysing performance relative to set goals.

Data Storage: For storing user preferences like theme selection (dark mode or light mode), I implemented Shared Preferences. This lightweight solution is ideal for saving simple key-value pairs, making it suitable for this option.

Navigation: To provide an organic and intuitive user experience, I included back buttons for straightforward navigation between adjacent screens. As well as this, I used data display elements as interactive navigation tools, allowing users to select a data card and navigate to its subtask or information page. This approach makes navigation seamless and user-friendly.

From the choice of development tools to the architectural components, were made with the dual goals of optimizing the app's functionality and ensuring a positive user experience. Each choice reflects a careful consideration of the requirements and the desire to create an app that is both powerful and easy to use.

Task

taskID	title	description	goalTime	completeTime	importance	stage
PK						

SubTask

subTaskID	taskID	title	description	goalTime	completeTime	importance	stage
PK	FK						

3. Novel Features

Time Goals for Tasks: A standout feature of Rootine is the ability to set time goals for each task. This functionality acts as a motivational tool by introducing faux deadlines, significantly enhancing an individual's ability to initiate tasks. It provides users with a clear timeframe for completion, providing a sense of urgency and improving time management skills.

Hierarchical Task Organization: Another unique aspect of Rootine is the capability to create tasks with nested subtasks. This hierarchical structure allows for organization and detailed planning, offering a level of depth not found in similar apps. It enables users to break down complex tasks into more manageable parts, providing a more systematic approach to task completion.

4. Challenges and Future Improvements

Designing the app's appearance was both the most enjoyable and time-consuming task. I spent a lot of time looking into colour theory, ensuring the use of complementary colours to make the text stand out.

However, managing the app's scope proved challenging. I had too many ideas for features that could enhance the app, and had to scale back due to time constraints, focusing on the core functionality first.

User Customization: Giving users more control over the app's appearance is important. The user could change the colour of the different priorities of tasks or change the order that they are sorted by.

Feature Expansion: Allowing the user to navigate to a webpage from a task could make the app more useful as well as a statistic page which would show interesting statistics like number of tasks completed this year.

Lastly: If I had more time I would have designed a logo for the app, this could be a root curling around for the logo